VASANTDADA PATIL PRATISHTHAN'S
COLLEGE OF ENGINEERING AND VISUAL ARTS
(An Autonomous Institute)
NAAC 'A' Grade Accredited | ISO 9001:2015 Certified
DEPARTMENT OF INFORMATION TECHNOLOGY
NBA Accredited (Dated 01/07/2024 to 30/06/2027)

CAMPUS NOTIFY: INSTITUTIONAL CIRCULAR & ANNOUNCEMENT MANAGEMENT SYSTEM

A Modular, Object-Oriented System in Core Java SE 17 with Responsive Web Interface

Review – 1 Presentation | Java Programming Laboratory (Semester – III)

Guide: Prof. Sonali Kharat

Team Details:

Student ID	Student Name
VU4F2526072	PRATIK ATPADKAR
VU4F2526066	HARSH TIWAR
VU4F2526062	PARTH GADADHE

Table of Content:

- Introduction
- Problem Statement
- Objectives
- Literature Survey
- Research Gap
- Proposed Solution
- Project Planning & Timeline
- Conclusion
- References

Introduction:

- **Institutional Context:** University campuses require fast, authenticated, and transparent dissemination of critical academic notices, examination timetables, fee circulars, and departmental events.
- **Limitations of Legacy Systems:** Traditional physical notice boards and informal WhatsApp broadcast groups lead to notice loss, cognitive clutter, and complete absence of delivery verification.
- **Need for Digital Centralization:** A single, authoritative digital notice board ensures official circulars reach every student, faculty member, and administrator without physical delays.
- **Core Java SE 17 Engineering:** Developed as part of the Java Programming Laboratory to demonstrate real-world applications of Object-Oriented Programming (OOP), file streams, and security models.
- **Target Stakeholders:** Specifically tailored for Department of Information Technology and administrative governance at Vasantdada Patil Pratishthan's College of Engineering & Visual Arts.

Problem Statement:

- **Physical Notice Board Bottlenecks:** Circulars get torn, misplaced, or buried under newer posters; students absent from campus miss urgent submission deadlines and academic announcements.
- **Social Media Chaos & Cognitive Fatigue:** WhatsApp and Telegram groups mix official circulars with informal chatter, memes, and non-academic media, leading to missed examination dates.
- **Zero Proof of Delivery & Acknowledgment:** Faculty and examination cells have no verifiable mechanism or digital logs to confirm whether targeted students actually read critical notices.
- **Vulnerability to Fake Circulars & Rumors:** Unofficial broadcast channels lack Role-Based Access Control (RBAC), leaving students susceptible to unverified exam cancellation rumors.
- **Paper Waste & High Operational Overhead:** High recurring paper printing costs, toner waste, and physical labor required to distribute and pin paper circulars across multiple campus buildings.

Objectives:

- **Centralized Digital Notice Repository:** To build a single authoritative portal categorized by department (IT, Comps, Mech, Civil) and priority (Urgent, Normal).
- **Implement Core Java OOP Architecture:** To apply Encapsulation, Inheritance, Polymorphism, and Interfaces (Notifiable, AuditLoggable) for robust, maintainable domain services.
- **Enforce Role-Based Access Control (RBAC):** To establish a Zero-Access Security Wall restricting circular creation, editing, and deletion strictly to authenticated Super Admins and Faculty.
- **Provide Verifiable Digital Acknowledgment:** To enable students to digitally acknowledge circulars with instant timestamped receipts for administrative audit records.
- **Deliver Responsive & Offline Accessibility:** To ensure 100% responsive user experience across mobile phones, tablets, and PCs with local caching during connectivity drops.
- **Promote Green Campus Sustainability:** To eliminate paper dependency, advancing UN Sustainable Development Goals (SDG 9: Innovation, SDG 12: Responsible Consumption).

Literature Survey:

Sr. No.	Title:	Publication	Author & Year	Methodology:	Key Findings:
1	**Digital Campus Information Dissemination Systems**	IEEE Access	R. Sharma et al. (2021)	Centralized push notification broker using message queues	High delivery rate across networks, but lacked role-based publishing controls and offline fallback.
2	**Smart Notice Board using IoT Cloud Services**	IJERT	P. Patel & S. Deshmukh (2022)	Raspberry Pi & LCD digital display displays	Reduced paper waste, but required expensive on-campus physical hardware; inaccessible remotely.
3	**Role-Based Information Management in Higher Ed**	Springer CS	A. Gupta & K. Rao (2023)	Multi-tier RBAC with relational access models	Proved security benefits of role boundaries, but lacked client-side caching and acknowledgment receipts.
4	**Campus Communication Bottlenecks in Higher Education**	Elsevier CIE	M. Fernandez et al. (2024)	Empirical survey across 12 university campuses	68% of surveyed students missed critical deadlines due to unverified social media broadcast groups.

Research Gap:

- **Lack of Strict Role-Based Publishing:** Existing communication channels either permit unrestricted student posting (leading to spam and rumors) or rely on complex enterprise ERPs that lack agile customization for college departments.
- **Absence of Verifiable Acknowledgment Receipts:** Current digital bulletin boards passively broadcast notices without recording timestamped student delivery acknowledgments, leaving faculty uncertain about message penetration.
- **Physical Hardware Dependency:** Existing smart notice boards require expensive LCD/LED hallway displays, failing to deliver notices directly into students' pockets on their personal mobile devices.
- **Zero Offline Fault Tolerance:** Most campus web portals fail completely when on-campus Wi-Fi disconnects; lack of local client-side caching prevents students from viewing previously loaded circulars.
- **Fragmented Priority & Category Routing:** Absence of structured priority filtering (Urgent vs. General) causes life-critical exam postponement alerts to be lost within routine club notices.

Proposed Solution:

- **Modular Core Java SE 17 Architecture:** Robust Model-View-Controller (MVC) design separating domain models (Notice, User), interfaces (Notifiable), and CLI / Web business logic.
- **Zero-Access Authentication & RBAC Gate:** Strict multi-tiered Role-Based Access Control ensuring circular creation, pinning, and deletion are restricted strictly to authenticated Super Admins and Faculty.
- **Dynamic Department & Priority Routing:** Instant categorization across departments (IT, Comps, Mech, Civil) and priority badges (Urgent, Normal) with sub-5ms render latency.
- **Verifiable Digital Acknowledgment Engine:** One-click student acknowledgment recording verified digital receipts for examination and placement compliance audit records.
- **Dual-Tier Persistent Storage:** Seamless integration combining real-time cloud data storage with local offline-first storage (localStorage) for uninterrupted offline resiliency.
- **Cross-Device Responsive Web Client:** Clean, responsive portal optimized for all smartphones, tablets, and desktop workstations with zero visual clipping or hidden actions.

Project Planning & Timeline:

Week / Duration	Phase Name	Key Tasks & Milestones	Status
Weeks 1–2 (July 2025)	**Need Identification & SRS**	Problem definition, literature survey, stakeholder requirement gathering	**Completed**
Weeks 3–4 (Aug 2025)	**Literature Survey & Design**	Comparative study, UML diagrams (Class, Use Case, Sequence), OOP modeling	**Completed**
Weeks 5–7 (Aug–Sep 2025)	**Core Java OOP Modules**	Implementing Notice, User, NotificationService, inheritance & interfaces	**Completed**
Weeks 8–9 (Sep 2025)	**Web Client & Cloud Sync**	Responsive dashboard, Firebase cloud sync, RBAC Zero-Access Wall	**Completed**
Weeks 10–11 (Oct 2025)	**Testing & Verification**	92 Automated test suite execution, stress testing, edge-case fixes	**Completed**
Weeks 12–13 (Oct 2025)	**Documentation & Review**	Report generation, logbook compilation, review presentation submission	**Completed**

Conclusion:

- **Successful Academic & Technical Delivery:** Campus Notify successfully bridges the gap between academic theory and practical software engineering using Core Java SE 17 object-oriented paradigms.
- **Elimination of Communication Bottlenecks:** Replaced fragmented physical notice boards and chaotic WhatsApp groups with a structured, instant, and searchable institutional platform.
- **Administrative Governance & Transparency:** Enforced strict Role-Based Access Control and verifiable digital acknowledgment receipts, completely eliminating lost circular disputes.
- **Environmental Sustainability (Green Campus):** Significantly reduces physical paper printing, toner consumption, and operational costs, aligning directly with UN Sustainable Development Goals (SDG 9 & SDG 12).
- **Future Scope & Roadmap:** Planned enhancements include native Android & iOS push notifications via Firebase FCM, automated SMS parent alerts, and direct integration with institutional biometric ERP databases.

References:

- [1] R. Sharma, K. Patel, and S. Deshmukh, “Digital Campus Information Dissemination Systems,” IEEE Access, vol. 9, pp. 11200–11210, 2021.
- [2] P. Patel and S. Deshmukh, “Smart Notice Board using IoT Cloud Services,” International Journal of Engineering Research & Technology (IJERT), vol. 11, no. 5, pp. 45–52, 2022.
- [3] A. Gupta and K. Rao, “Role-Based Information Management in Higher Education,” Springer Computer Science, vol. 48, pp. 210–225, 2023.
- [4] M. Fernandez, L. Gomez, and T. Silva, “Campus Communication Bottlenecks in Higher Education,” Elsevier Computers in Industry & Education, vol. 38, pp. 102–115, 2024.
- [5] H. Schildt, Java: The Complete Reference, 12th ed., McGraw-Hill Education, 2021.
- [6] J. Bloch, Effective Java, 3rd ed., Addison-Wesley Professional, 2018.
- [7] University of Mumbai, Syllabus for Second Year Information Technology (Semester III & IV), 2024–2025.
- [8] United Nations, Sustainable Development Goals Knowledge Platform, [Online]. Available: https://sdgs.un.org/goals.

VASANTDADA PATIL PRATISHTHAN'S
COLLEGE OF ENGINEERING AND VISUAL ARTS
(An Autonomous Institute)
NAAC 'A' Grade Accredited | ISO 9001:2015 Certified
DEPARTMENT OF INFORMATION TECHNOLOGY
NBA Accredited (Dated 01/07/2024 to 30/06/2027)

Thank You!

Any Questions or Suggestions?

Team Members: Pratik Atpadkar (VU4F2526072) | Harsh Tiwar (VU4F2526066) | Parth Gadadhe (VU4F2526062)

Project Guide: Prof. Sonali Kharat | **Head of Department:** Dr. Pradip Mane

Department of Information Technology

Vasantdada Patil Pratishthan's College of Engineering & Visual Arts, Sion, Mumbai

Live Portal: **https://campnotify-eef91.web.app**